HAWASSA UNIVERSITY

INSTITUTE OF TECHNOLOGY

FACULTY OF INFORMATICS

DEPARTMENT OF COMPUTER SCIENCE

Host Company: INSA(Information Network Security Agency)

Duration of the practical attachment: 3 months

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Declaration

We declare that this internship report paper is submitted to Hawassa University, Computer Science Department. It is a record of original work done during the three-month internship period at the Information Network Security Agency (INSA), Data Analysis Department. This paper is written through our own effort to briefly explain our experiences and activities during the internship period.

Acknowlagment

We, the undersigned interns, would like to express our sincere gratitude to the **Information Network Security Agency (INSA)** for granting us the opportunity to undertake our practical attachment within its **Data Analysis Department**. The organization provided us with a rich learning environment, resources, and mentorship that made this experience truly rewarding.

We extend our heartfelt thanks to our supervisors and mentors at INSA for their consistent guidance, technical support, and constructive feedback throughout our projects. Their mentorship enabled us to successfully contribute to the tasks assigned while enhancing our knowledge and skills.

Our appreciation also goes to our university, faculty, and department for facilitating the attachment program and ensuring that we gained practical exposure to complement our academic studies. Finally, we would like to acknowledge our family and friends for their encouragement and moral support throughout this journey.

Executive Summary

This report documents three advanced projects completed over the summer: **Image Forgery Detection, Sentiment Analysis, and Keypoint Detection.** Together, these projects demonstrate the practical application of artificial intelligence across computer vision and natural language processing, highlighting how deep learning can be used to address pressing real-world challenges.

The **Image Forgery Detection project** aimed to identify manipulated or fake images, particularly deepfakes, which threaten the credibility of digital content. Using deep convolutional neural networks and models such as F3Net, the system was trained to detect pixel-level and frequency-based inconsistencies that are often invisible to the human eye. This work illustrates the role of AI in strengthening digital security, protecting individuals and institutions against misinformation and fraud.

The **Sentiment Analysis project** concentrated on natural language processing, with a focus on Amharic and multilingual text data. By fine-tuning transformer-based architectures like XLM-RoBERTa, the system successfully classified text into sentiment categories such as positive, negative, or neutral. This provides an essential tool for analyzing public opinion, enabling businesses, policymakers, and researchers to make informed decisions based on real-time language data.

The **Keypoint Detection project** explored the use of deep learning to automatically identify critical points in images, such as facial landmarks or body joints. Leveraging CNN-based architectures, the project demonstrated the ability of AI to interpret visual structures, with applications in healthcare (e.g., movement analysis), human-computer interaction, augmented reality, and security systems.

Collectively, these projects underscore the importance of **data-driven innovation** and showcase the skills developed in handling datasets, implementing neural architectures, and evaluating results using standard metrics. Beyond the technical implementation, the projects emphasize how AI can enhance **trust, communication, and human-centered technologies**, while also identifying areas for future research, including larger datasets, optimization for real-time applications, and improved model interpretability.

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Chapter One: Background of INSA

1.1 Introduction

The **Information Network Security Agency (INSA)** is a government organization in Ethiopia that focuses on protecting the country's information, computer systems, and networks. In today's world, technology is part of almost everything we do. It is used for communication, banking, education, business, healthcare, and government services. While technology brings many opportunities, it also comes with serious risks like hacking, data theft, malware, and different types of cyber attacks. If these threats are not controlled, they can cause a lot of damage to national security, the economy, and people's daily lives.

Because of these challenges, INSA was established to defend Ethiopia against cyber threats and to make sure the country's information systems are safe, reliable, and trustworthy. INSA's work is not only about stopping attacks but also about preparing for the future. The agency is involved in **research, innovation, and capacity building**, which means training skilled people and developing local knowledge in important areas such as data analysis, software development, artificial intelligence, and cryptography. In short, INSA is both a **protector** of Ethiopia's digital world and a **builder** of the nation's future in technology.

1.2 Brief History of INSA

INSA was officially established in **2006 (1999 E.C.)** by the Ethiopian government. At first, the agency started with only a few experts and its main mission was simple: to protect government offices and information systems from cyber attacks. Over the years, however, the role of INSA expanded as Ethiopia's use of technology grew.

When more people began to access the internet and digital services started spreading across the country, new challenges appeared. Cyber attacks became more common and more dangerous. Because of this, INSA took on bigger responsibilities such as developing national cyber infrastructure, creating cybersecurity policies, and training professionals in the field.

Today, INSA is recognized as a key institution in Ethiopia's digital transformation. It not only works with government offices but also supports banks, schools, health institutions, defense, and private companies.

Figure 1.1: Logo of INSA

1.3 Main Product and Services of INSA

The Information Network Security Agency (INSA) is Ethiopia's primary institution responsible for ensuring national cybersecurity, information safety, and digital sovereignty. Over the years, it has become a strategic center for cybersecurity research, policy development, and applied technology solutions. Some of its main products and services include:

Cybersecurity and Information Protection

One of INSA's major responsibilities is protecting Ethiopia's digital space. The agency makes sure that sensitive information and national networks are kept safe from cyberattacks. This includes blocking hackers, stopping harmful software, and protecting against digital spying. It also creates strong defense strategies to ensure that government systems continue to work smoothly and securely.

Data Analysis and Intelligence

Another important service of INSA is studying and understanding large amounts of information. By carefully analyzing data, the agency can provide valuable insights that help the government make better decisions. Some examples include opinion mining (studying public views), text analysis, and checking whether digital images are original or manipulated. In today's world, where information spreads fast, these services are very useful.

Digital Forensics

When cybercrimes happen, INSA acts as an investigator. It can examine computers, mobile devices, and online activity to find out how a crime was committed. The agency collects and studies digital evidence and supports the justice system in identifying and prosecuting cybercriminals.

Policy and Capacity Building

Apart from technical work, INSA also develops national policies that guide how the country should protect its digital infrastructure. At the same time, it trains experts in fields like ethical hacking, secure data handling, and digital investigations. This helps Ethiopia to grow its own skilled professionals who can handle future cyber challenges.

Software Development Team

INSA has dedicated teams that design and build software solutions for both government and public services. These systems are developed to be safe, reliable, and suitable for Ethiopia's needs. Some examples include secure communication platforms, data management systems, and applications that support government operations.

Penetration Testing

To ensure that systems are safe, INSA performs penetration testing. This is when specialists intentionally test networks and software to see if there are weaknesses. By finding these weaknesses before criminals do, the agency helps organizations strengthen their defenses.

Malware Detection and Response

INSA also works to identify and remove harmful software that may attack systems. Its teams monitor networks and devices for viruses, worms, and other types of malware. Once found, they create solutions to clean and protect the systems from future threats.

Data Center Services

As part of its infrastructure, INSA manages secure data centers that host and protect sensitive government information. These centers provide storage, backup, and reliable access to data while ensuring that security standards are maintained.

Fayda Digital ID

INSA also plays a role in supporting Fayda, Ethiopia's digital ID system. Fayda provides citizens with a secure and trusted digital identity that can be used for different services. It helps improve service delivery, reduces fraud, and makes it easier for people to access both public and private sector services.

In general, INSA's products and services are not limited to just cybersecurity but extend to applied AI research, data intelligence, and capacity building. Thus, it is the most technologically advanced government agencies in Ethiopia.

1.4 Main Customers and End Users of INSA’s Products and Services

INSA’s customers are diverse, ranging from **high-level state institutions** to **ordinary citizens** who benefit indirectly from its protective and technological services. Below is a breakdown of its primary customer groups and how they engage with INSA’s products and solutions:

1. Government Institutions

Government offices represent one of INSA’s most important customer bases. Ministries, federal agencies, and regional authorities rely on INSA to protect sensitive data, secure communication systems, and enable safe digital platforms.

- **Key Needs:**
  - Protection of sensitive government records (diplomatic files, national databases, budgetary data).
  - Secure communication channels between officials.
  - E-government services free from breaches or disruptions.
- **Examples:**
  - Ministry of Finance → secure online revenue collection systems.
  - Ministry of Health → safe storage of patient data and medical research.
  - Ministry of Education → secure digital examination systems and academic record databases.

Without INSA’s services, these institutions would face constant risks of data theft, hacking, and loss of credibility in the delivery of public services.

2. Critical Infrastructure Providers

INSA plays a central role in protecting Ethiopia’s **critical infrastructure** sectors. These are industries that, if disrupted, could paralyze the country’s economy and daily life.

- **Banking and Financial Sector:**
  - Protection of mobile banking platforms, ATMs, and online transactions.
  - Fraud detection systems to prevent theft of customer funds.
  - Compliance with global cybersecurity standards to build public trust.
- **Telecommunications Sector:**
  - Safeguarding the backbone of Ethiopia’s communication networks.

  - Preventing attacks on mobile networks and internet services.
  - Ensuring resilience of EthioTelecom and new service providers.
- **Energy and Transport:**
  - Security for electric power grids and hydroelectric plants.
  - Protection of aviation systems from digital sabotage.
  - Safe operation of logistics and transport management systems.

For these sectors, INSA's role is not optional but essential, as a single cyber incident could have **nationwide consequences** affecting millions of citizens.

3. Defense and National Security Sector

The military and intelligence community represent a core end user of INSA's expertise. Secure communication, surveillance protection, and cyber defense systems developed by INSA are indispensable for safeguarding Ethiopia's sovereignty.

- **Services Provided:**
  - Encrypted communication platforms for defense operations.
  - Threat intelligence systems that monitor and prevent cyber warfare.
  - Digital forensics support to investigate and neutralize cyber incidents.

INSA functions as the **technological backbone of national security**, ensuring that Ethiopia's defense institutions are not vulnerable to digital infiltration or espionage.

4. Research, Academic, and Innovation Communities

As Ethiopia seeks to become a digital economy, INSA also supports knowledge creation and academic collaboration. Universities, research centers, and startups engage with INSA for:

- **Capacity building:** training students, researchers, and ICT professionals in cyber defense and AI.
- **Collaboration:** joint projects on AI, data science, cryptography, and robotics.
- **Technology hosting:** providing secure environments for innovation and experimentation.

This makes INSA both a **protector and a partner** in Ethiopia's pursuit of technological sovereignty.

5. The General Public (Indirect End Users)

- Although ordinary citizens are not direct clients of INSA, they remain the ultimate beneficiaries of its services. By securing national systems and critical infrastructure, INSA ensures that Ethiopians enjoy:
- **Safe e-government platforms** for tax filing, ID services, and digital applications.
- **Reliable communication networks** free from frequent outages or breaches.
- **Protection against cybercrime** such as phishing, fraud, and online identity theft.
- **Trust in digital services** that underpin financial transactions, education, and healthcare.

In this way, every Ethiopian who uses mobile banking, accesses online education, or engages with government services indirectly depends on INSA's work.

Summary of Customer Base

In summary, INSA's main customers and end users can be grouped as follows:

1. **Primary Customers (Direct):**
    - Government ministries and agencies.
    - Critical infrastructure operators (finance, telecom, energy, transport).
    - Defense and security institutions.
2. **Secondary Customers (Indirect):**
    - Research and academic institutions.
    - Private sector innovators and startups.
    - The general public (through protected services).

By serving these groups, INSA ensures not only the **protection of national security** but also the **stability of Ethiopia's digital transformation**. Its role as both a security provider and technology enabler makes it one of the most significant institutions driving the country's future.

1.5 Organization and Structure

INSA is a large institution with different departments and divisions. Each department has its own focus, but they all work together to achieve the same mission. Some of the main departments are:

- **Cybersecurity Department** – protects networks and responds to cyber attacks.
- **Data Analysis Department** – analyzes big data to find useful information and support decision-making. (This is where we worked during our internship.)
- **Research and Development Department** – focuses on innovation and creating new technologies.
- **Training and Capacity Building Department** – organizes training programs for professionals, students, and partner institutions.
- **Administration and Support Department** – manages staff, resources, and logistics.

The **Data Analysis Department**, where we worked, is one of the newer departments. Its main responsibility is to process large amounts of data, identify patterns, and produce insights that are helpful for national security and government planning.

1.6 Work Flow of INSA

The workflow of INSA is well-organized and follows a systematic approach. For example:

1. A cyber threat or data-related issue is first identified by the monitoring team.
2. The task is sent to the relevant department.
3. The department forms a team of experts to investigate the issue.
4. The team collects data, performs analysis, and designs possible solutions.
5. The results are reported to INSA management and, when necessary, shared with government authorities.
6. If required, INSA also provides training or awareness programs to other institutions so that similar problems can be prevented in the future.

Chapter Two: Our Practical Attachment Experience

2.1 How We Get Into The Company

Our team joined INSA through the university's Practical Attachment program, which is designed to connect students with organizations that match their academic field. After submitting the application letter from our university, INSA responded within two days and invited us to report to their office.

When we arrived, we were welcomed with an orientation program that introduced us to the agency's structure, core responsibilities, and different departments. Each department presented its role. For example, the penetration testing team explained how they simulate attacks to check system weaknesses, while others described areas like malware detection, and software development.

Among these, we were interested to the Data Analysis Department because of its strong connection to artificial intelligence and machine learning.

2.1.1 Placement within INSA: Data Analytics in AI and ML

For a period of three months, we were placed within the Data Analytics Division of the Information Network Security Agency (INSA), working specifically in the streams of Artificial Intelligence (AI) and Machine Learning (ML). This division represents one of the most innovative and strategically important units of INSA, tasked with transforming raw, complex datasets into actionable intelligence that directly supports national security, policy formulation, and technological advancement.

The AI and ML streams function as the analytical backbone of INSA's operations. Their mandate is not only to secure Ethiopia's critical digital infrastructure but also to drive forward research and development in emerging technologies. The division achieves this by:

- Developing intelligent models capable of detecting anomalies, classifying high-volume datasets, and automating complex decision-making.
- Advancing computer vision applications, such as forgery detection, biometric verification, and keypoint recognition, to reinforce the credibility of digital systems.
- Building natural language processing (NLP) systems for sentiment detection, multilingual analysis, and information monitoring to assist government and policy stakeholders.
- Applying predictive analytics to anticipate risks and support strategic planning for government institutions, financial sectors, and defense operations.

These streams are especially significant within INSA because they bridge the gap between data and decision-making. By harnessing AI-driven analytics, the division ensures that Ethiopia can identify risks early, respond to threats effectively, and pursue digital innovation with sovereignty and confidence.

Supervision and Research Leadership
Our three-month engagement was conducted under the close supervision of two highly respected experts within INSA:

- **Mr. Tewodros Legesse**, Senior Data Analytics Researcher, who provided overall guidance on the direction of our division's work. His expertise and leadership ensured that our research aligned with INSA's national mission while maintaining technical rigor.
- **Dr. Seble Hailu**, under whose mentorship we worked on classification-based machine learning projects. She guided us through the application of supervised learning techniques, ensuring methodological precision and relevance to INSA's applied research goals.

Through their mentorship, we gained not only technical expertise in applying deep learning and machine learning methods but also practical insight into how these technologies directly serve Ethiopia's national priorities.

Alignment with INSA's Mission
Our contributions over these three months centered on three projects—image forgery detection, sentiment analysis, and keypoint detection. Each project represented a different application of AI within the Data Analytics streams but shared a common purpose: to strengthen Ethiopia's capacity in cybersecurity, information integrity, and human-centered technologies. These projects demonstrated how research at the division level is translated into solutions with national impact, serving INSA's diverse stakeholders, from government institutions and critical infrastructure providers to the general public.

2.2 Workflow in the Data Analysis Department

The Data Analysis Department has several sub-groups working on different projects. Each group has a leader who checks the progress of the project. The leaders report how far the work has gone and what challenges they face.

we did not have full information about daily routines or meetings, but from our observation, the groups are organized, and tasks are assigned based on project needs.our group was working on different project under the guidance of our supervisor.

2.3 Work Tasks Executed During the Practical Attachment

During our practical attachment at INSA, our group worked on three major projects: Image Forgery Detection, Opinion Mining, and Key Point Detection. Each project was carefully introduced to us, and before we began any practical coding, we spent time building a strong understanding of the topics.

Reading Research Papers and Preparing for Implementation

One of the first things we did after receiving our project assignments was to study research papers and articles related to each project. This step was very important because it gave us a solid background and helped us make informed decisions about the direction of our work. Through this reading, we were able to:

- Understand the basic concepts behind each project, such as how deep learning detects forged images, how natural language processing extracts meaning from text, and how computer vision identifies key points in images.
- Compare different models that researchers had used for similar problems. We looked at their strengths and weaknesses, the datasets they used, and how well they performed based on reported results.
- Choose appropriate models for our projects by seeing which approaches worked best in research and matched our available resources.
- Evaluate datasets by learning which datasets were commonly used in studies and how they were structured (e.g., labeled images, annotated key points, or text sentiment labels).
- Learn about metrics (accuracy, precision, recall, F1-score, AUC, etc.) that are used to evaluate models, so we could later measure how well our implementations were performing.

This preparatory step ensured that when we started working on the projects, we already had a strong idea of what to expect, what tools to use, and how to measure success. It also helped us save time, since we were not blindly experimenting but instead working with a roadmap inspired by existing research.

2.3.1 Image Forgery Detection Project

This project was one of the most exciting and technically challenging tasks we worked on. The main objective was to develop a system that could identify manipulated or fake images, which is a growing problem in today's digital world. Our key tasks included:

Understanding how F3Net model:
At the beginning, we spent time learning about the F3Net deep learning model, which is specifically designed for image forgery detection. We went through research papers and tutorials to understand how the model processes images and distinguishes between real and fake ones.

Preparing the dataset:
We were given a large collection of images, but they were not immediately ready for training. As a group, we cleaned the dataset by removing duplicate images, organizing them into proper folders, and ensuring the images were in the correct format. This step was very important because the quality of the data directly affects the accuracy of the model. And then split 80% of the dataset for training, 10% for validation and 10% for testing. This is very helpful for evaluating the final model's performance on unseen data.

Training and testing the model:
After preparing the dataset, we trained the F3Net model to recognize forgery patterns. This required running multiple experiments, adjusting parameters, and monitoring the training process. Later, we tested the model on new images to evaluate its accuracy.

Debugging and optimization:
While working with the model, we faced several errors and challenges, especially with compatibility issues and accuracy. As a group, we collaborated to debug these errors, improve the training process, and ensure the model produced reliable results. And with a lot of training, we have achieved 94% AUC and 0.00004 loss.

Through this project, we gained strong technical skills in **deep learning, dataset management, model training, and collaborative debugging**.

2.3.2. Opinion Mining Project

Our second project focused on Opinion Mining, also known as Sentiment Analysis, where we analyzed Amharic text datasets to identify whether the sentiment expressed in the text was positive, negative, or neutral. Our key tasks included**:**

Data cleaning and preparation:
The raw dataset contained a lot of noise such as duplicates, unnecessary symbols, and irrelevant characters. We cleaned the text by removing these issues so that the data could be effectively processed.

Natural Language Processing (NLP) techniques:
Since the dataset was in Amharic, we applied NLP techniques like tokenization (splitting sentences into words), stemming (reducing words to their base form), and stop-word removal (eliminating common but unimportant words). This step helped the computer better understand the structure and meaning of the text.

Model training and classification:
We then trained XLM-ROBERTA model to classify the text into three categories: positive, negative, or neutral. This required experimenting with different algorithms and evaluating their performance.

Visualization and reporting:
After training, we visualized the results using charts and graphs, which made it easier to understand the distribution of sentiments in the dataset. Finally, we documented our findings, challenges, and solutions for future reference.

Working on this project improved our understanding of text analytics, machine learning for natural languages, and data visualization techniques. It also showed us the challenges of working with local languages like Amharic, where tools and resources are not as developed as for English.

2.3.3 Human Keypoint Detection Project

Our third project was about Human Keypoint Detection. In this project, we worked on detecting 17 keypoints of the human body using the COCO dataset. These keypoints represent important joints like shoulders, elbows, knees, and ankles. After detecting them, the system connects the points to draw a skeleton of the person. This type of technology can be used in many areas, such as security monitoring, sports performance analysis, and human-computer interaction.

We started by learning about the dataset and the models we could use. The COCO dataset contains images with people where each body joint is already labeled. We also read research papers to understand how keypoint detection works and why certain models are better. For our system, we used HRNet and Lite-HRNet as backbone models, and we added Faster R-CNN to detect multiple people in one image. After detecting people, the system cropped them and then passed them to the keypoint detector, which marked the joints and connected them into skeletons.

Our main tasks were guided by our supervisor. First, we debugged and understood the given code in PyCharm. Then we made changes so the models could detect all 17 keypoints. We also prepared PowerPoint presentations to explain our progress to our mentor and answered questions after the presentations. For training, we used 1000 images from the COCO dataset and trained the model for 10 epochs on Kaggle and

Google Colab. For larger training, our supervisor ran the experiments on INSA's servers to get better results.

This project gave us hands-on experience with tools like PyCharm, Python, PyTorch, Kaggle, and Google Colab. We learned how to combine backbone models with object detectors, how to prepare datasets, and how to test and improve models. Most importantly, it helped us understand how human keypoint detection can be applied in real systems and the challenges that come with training and debugging deep learning models.

2.5 Procedures and Methodology

Over the course of our **three-month placement** within the **Data Analytics Division (AI & ML stream) at INSA**, our workflow was designed to combine **theoretical study, research review, technical implementation, and applied system development.** This ensured that we did not simply experiment with models in isolation but followed a disciplined process that reflects best practices in research-driven AI projects.

The process unfolded in the following stages:

1. Studying and Revising Machine Learning Foundations

Our work began by consolidating our knowledge of **machine learning (ML) fundamentals**, ensuring that all members of the team shared a common technical foundation. This stage included:

- Revisiting core ML concepts: supervised vs. unsupervised learning, classification, regression, and clustering.
- Studying **deep learning architectures** such as CNNs (for image-based tasks), RNNs/LSTMs (for sequential tasks), and Transformers (for NLP tasks).
- Reviewing evaluation metrics: accuracy, precision, recall, F1-score, PSNR, SSIM, and confusion matrices.
- Conducting internal discussions to connect these concepts with the **practical security challenges** INSA addresses.

This early stage gave us the grounding necessary to critically analyze research papers and implement models later with confidence.

2. Research Review and Knowledge Exchange

The second stage involved **academic research and peer learning**. We reviewed and analyzed **state-of-the-art research papers and technical reports** related to our project domains:

- **Image Forgery Detection:** frequency-based networks, pixel inconsistency detection, and models such as F3Net.
- **Sentiment Analysis:** NLP models with emphasis on multilingual embeddings, particularly XLM-RoBERTa.
- **Keypoint Detection:** CNN-based landmark detection methods, applied in computer vision and biometric analysis.

Each member prepared **PowerPoint presentations** summarizing their assigned papers. These sessions served as **mini research seminars**, allowing us to:

- Exchange knowledge across project domains.
- Discuss strengths and limitations of each model.
- Identify gaps between academic research and INSA's real-world application needs.

Through this, we developed the ability to **critically evaluate AI research** and align our work with practical national priorities.

3. Model Selection and Justification

Following the research phase, we conducted a **comparative assessment** to select the most appropriate models for each project. Selection was based on three criteria:

1. **Accuracy and Reliability** – ensuring robustness in detecting forgeries, classifying sentiment, or identifying keypoints.
2. **Computational Efficiency** – selecting models that balance performance with available hardware resources.
3. **Applicability to INSA's mission** – prioritizing solutions that address Ethiopia's security and information integrity needs.

Final choices included:

- **F3Net** for forgery detection (due to its strong ability to capture pixel-level inconsistencies).
- **XLM-RoBERTa** for sentiment analysis (for its multilingual representation power).
- **CNN-based landmark detection models** for keypoint tasks.

This phase highlighted the importance of **decision-making grounded in evidence**, not just code execution.

4. Code Implementation and Execution

- Once models were selected, we proceeded to **practical implementation** using Python and frameworks such as **PyTorch and TensorFlow**. This included:
- Setting up environments in **PyCharm and Kaggle**, with Linux-based configurations for efficiency.
- Preprocessing datasets: cleaning, normalizing, and preparing data pipelines for training.
- Writing modular scripts for training, validation, and testing to ensure reusability.
- Structuring the code so that experiments could be replicated or extended in the future.

This stage transformed research into a **working system**, bridging the gap between theory and application.

5. Debugging and Optimization

During implementation, we encountered challenges such as dataset path mismatches, version conflicts between libraries, and unstable training behavior. We applied systematic **debugging strategies**, including:

- Line-by-line error tracing to identify and fix bugs.
- Optimizing code structure for computational efficiency.
- Iteratively testing and adjusting preprocessing steps to ensure data integrity.

By refining our scripts, we arrived at the **best possible version of the code**, one that was both stable and efficient for model training.

6. Model Training and Evaluation

With stable code in place, we trained our chosen models on curated datasets. This stage required balancing **technical detail with practical constraints**:

- Adjusting hyperparameters (learning rate, batch size, epochs).
- Using metrics suited to each project: e.g., SSIM/PSNR for forgery detection, accuracy/F1-score for sentiment analysis, and precision of detected landmarks for keypoint tasks.
- Monitoring loss curves and validation accuracy to avoid overfitting.
- Recording and analyzing results systematically to inform further adjustments.

This iterative process of **training → evaluation → refinement** allowed us to achieve the **best performance possible within the available resources.**

7. User Interface Development and Demonstration

To make our work accessible to both **technical experts and non-technical stakeholders**, we developed **user interfaces (UIs)** for each project. These UIs allowed for live demonstrations, such as:

- Uploading an image to check for manipulations.
- Entering or pasting text to classify its sentiment.
- Providing an input image or video frame for automatic keypoint detection.

The UIs transformed our research from abstract models into **practical, interactive tools**, showcasing the potential of AI solutions for INSA's mission and for Ethiopia's broader digital transformation.

8. Presentation and Knowledge Sharing

At the conclusion of each phase, we prepared **structured presentations** of our work, including:

- Summarized research findings.
- Demonstrations of implemented models.
- Insights from debugging and training.
- Future improvement directions.

These presentations allowed supervisors and peers to evaluate our progress and ensured transparency in our methodology.

Summary of our Workflow

Our overall workflow followed a systematic path:

Study --- Research Review--- Model Selection--- Code Implementation---- Debugging ---- Model Training -- UI Development --- Presentation & Demo.

This structured approach ensured that our engagement went beyond mere experimentation. Instead, we demonstrated the ability to:

- Understand ML theory.
- Critically assess existing research.

- Implement and refine state-of-the-art models.
- Train and evaluate models rigorously.
- Build UIs to communicate our results effectively.

This process reflects not only technical competence but also the **discipline and research methodology expected within a national-level institution like INSA.**

2.6 How Good We Have Been in Performing Our Work Tasks

When we started our internship, many things were new for us. At first, some tasks looked very difficult, but slowly we learned how to do them. We tried to understand every task our supervisor gave us. We listened carefully and asked questions when we did not understand. Because of this, we could understand tasks more easily after some days. Most of the time, we completed our work on time.

Our supervisor and often said we were doing good work. He told us we learned well and tried to do our tasks in the correct way. Sometimes he corrected our mistakes, but he also encouraged us and said we were improving every day. This gave us more energy to work better.

We also feel that we became better day by day. At first, we were not very confident, but later we felt more capable. We could do tasks without much help. We saw our progress in many small things. For example, when we debugged code or prepared reports, at first it took us a long time, but later we could do it more quickly and correctly.

During our internship, we showed many good qualities. We tried to be responsible and serious with our tasks. We always came on time and tried not to waste time. We were curious and wanted to learn new things, so we asked questions and read more when needed. We worked together with other people, respected them, and shared ideas. When problems came, we tried to solve them ourselves instead of waiting for others.

Some examples of our good work include: debugging code that was not working before, helping to prepare simple reports, analyzing data and presenting results. In these tasks, we tried our best, and the results were good.

overall, we can say we performed our internship tasks well. We learned a lot, improved our skills, and completed real work that helped the section and our team.

2.7 Challenges Faced During the Practical Attachment

During our time at INSA, we faced several challenges while working on the assigned projects. These challenges came from both technical and organizational aspects, and they tested our ability to adapt, learn, and work together as a team.

1. Technical Challenges

Compatibility issues with models and libraries:
Setting up deep learning models such as F3Net required multiple libraries and dependencies. Often, some versions of Python packages did not work well together, which caused errors and delayed our progress.

PC performance limitations:
The computers we used sometimes struggled to handle the heavy requirements of PyCharm and its environment. As a result, the overall performance slowed down, and this made training deep learning models even more time-consuming.

Storage space shortages:
Since datasets for image and text analysis are usually very large, we faced difficulties downloading and storing them on the available machines. This limited our ability to experiment with bigger and more diverse datasets.

Debugging deep learning models:
Errors were common during training and testing. Some of them came from the code, while others were related to GPU usage, environment setup, or even file paths. Debugging these issues required patience and teamwork.

2. Resource Limitations

Insufficient computing power:
Model training took a very long time because the available computers could not handle large-scale processing efficiently. This meant that some experiments had to be run overnight or repeated after crashes.

File transfer difficulties:
Moving files between servers and local machines was another challenge. Sometimes, dataset transfers were interrupted or corrupted, which forced us to restart the process and lose valuable time.

3. Time Management and Learning Curve

Being new to such projects:
Since this was our first time working on large-scale deep learning and data analysis

projects, it took us a while to understand the concepts, tools, and workflow. Adjusting to the professional standards at INSA required extra effort.

Research-heavy beginning:
For the first two months, we focused on reading research papers to understand the basics of forgery detection, opinion mining, and key point detection. While this gave us a strong foundation, it also consumed a lot of time.

Tight project schedule:
After spending two months on background reading, we had only about one month left to implement and test all three projects. This made us work under pressure and in a hurry, even though projects of this type usually require a longer timeline.

4. Team Coordination

Overlapping responsibilities:
With three members in the group, sometimes our tasks overlapped, or we repeated work unknowingly. This taught us the importance of clear communication and proper planning.

Balancing workload:
Since each project required multiple steps (data preparation, model training, debugging, documentation), dividing work fairly and ensuring everyone contributed equally was sometimes difficult.

2.8 Measures Taken to Overcome Challenges

Despite the many difficulties we encountered during our three-month attachment at INSA, we managed to overcome them through a combination of **technical strategies,**

resource management, teamwork, and personal discipline. These measures not only helped us progress with our projects but also taught us how to adapt to real-world working conditions.

1. Addressing Technical Challenges

a. Compatibility issues with models and libraries

- We carefully reviewed documentation for each library and experimented with different versions until we found stable combinations that worked with our models.
- Online resources such as GitHub issue pages, Stack Overflow discussions, and official PyTorch/TensorFlow guides were used to troubleshoot errors.
- We standardized our environments within the team to ensure everyone was working with the same versions, which reduced repeated problems.

b. PC performance limitations

- We optimized our code by using smaller batch sizes and lighter configurations during testing phases before moving to full-scale training.
- Wherever possible, we used pre-trained models and transfer learning techniques, which required less computational power compared to training from scratch.
- To avoid system slowdowns, we closed unnecessary background applications and used modular scripts to reduce memory consumption.

c. Storage space shortages

- We compressed large datasets into smaller manageable files and used only the most relevant subsets for training.
- When possible, we stored datasets on shared drives and external storage devices to reduce pressure on local machines.
- Data cleaning steps were applied early, removing duplicates and irrelevant files to save space.

d. Debugging deep learning models

- We approached debugging systematically: reproducing errors step by step, isolating issues, and testing in smaller chunks of code before running the full pipeline.
- We adopted pair-programming within the team, where two members worked together on difficult bugs while the third documented changes.

- We created logs and used error-tracking comments in our scripts so that recurring issues could be resolved faster in the future.

2. Managing Resource Limitations

a. Insufficient computing power

- To handle long training times, we scheduled experiments to run overnight and monitored outputs the next morning.
- We relied on checkpoints and saving intermediate results, which allowed us to resume training without restarting from zero after system interruptions.
- For certain tests, we used smaller datasets or reduced image sizes during experimentation, and only scaled up once the models were stable.

b. File transfer difficulties

- We adopted compressed file formats (e.g., .zip, .tar.gz) to speed up transfers and reduce the risk of corrupted files.
- In case of interruptions, we used segmented downloading tools and resume options to continue from where transfers stopped.
- Team members coordinated to share datasets directly within the group, so that one successful download could serve everyone.

3. Overcoming Time Management and Learning Curve Issues

a. Being new to such projects

- We relied heavily on mentorship from **Mr. Tewodros Legesse** and **Dr. Seble Hailu**, who guided us on prioritizing tasks and focusing on essential aspects rather than trying to cover everything at once.
- We organized peer-learning sessions where each member taught the others about specific models or techniques they had studied. This not only saved time but deepened our collective understanding.

b. Research-heavy beginning

- To maximize the usefulness of our first two months of reading, we prepared **PowerPoint presentations** summarizing each paper, ensuring that the time invested turned into shared group knowledge.
- We extracted key methods and results from papers to directly inform our coding decisions, so the reading was not wasted effort but an essential foundation.

c. Tight project schedule

- Once we moved into the coding stage, we shifted to a **task-priority system**, focusing on implementing the core functionality of each project first, and leaving extra features for later.
- We divided projects into smaller milestones, so progress could be measured weekly.
- By setting personal deadlines earlier than the official ones, we created buffers to handle unexpected delays.

4. Improving Team Coordination

a. Overlapping responsibilities

- We introduced a **clear division of tasks** by assigning one member as the lead for each project (forgery detection, sentiment analysis, and keypoint detection). This ensured ownership and reduced duplication.
- Weekly coordination meetings were held to update each other on progress, identify overlaps, and reassign tasks if necessary.

b. Balancing workload

- We rotated responsibilities so that no member was overloaded with repetitive tasks like debugging or documentation.
- Shared tools (Google Docs, version control for code) were used to track contributions transparently, ensuring fairness.
- Open discussions were encouraged whenever someone felt overwhelmed, so adjustments could be made in real time.

Summary of the Measures that we take

In short, we overcame our challenges by:

- **Adapting technically** through debugging, optimization, and environment management.
- **Managing resources wisely** by using smaller datasets, checkpoints, and compressed file formats.
- **Improving our time use** through mentorship, peer learning, and milestone planning.

- **Strengthening teamwork** with clearer roles, better communication, and fair workload distribution.

Through these measures, we turned challenges into learning opportunities. Instead of blocking our progress, the obstacles helped us grow in **problem-solving, adaptability, and collaboration** — skills just as valuable as the technical knowledge we gained.

Chapter Three: Benefits we Gained

3.1 Practical Skills

During our three-month internship, we were able to significantly improve our practical skills, especially in areas directly related to programming, debugging, and machine learning workflows.

One of the most important achievements was strengthening our Python programming skills. Although we don't had that much prior knowledge of Python from our university courses, we learn it quickly and by working on real-world projects at the internship allowed us to deepen our understanding and apply it in practice. We learned how to write clean, efficient code that could be reused and maintained easily. We also gained more experience in handling different Python libraries, especially those commonly used in data analysis and machine learning tasks.

Another key area of growth was debugging code using PyCharm. At the beginning of the internship, we often struggled when errors occurred in our programs. However, by consistently using the debugging tools provided by PyCharm, we learned how to carefully trace code execution, identify bugs, and fix them effectively. This helped us build confidence in troubleshooting problems independently and improved our overall problem-solving abilities.

Additionally, we acquired valuable experience in training machine learning models using Kaggle and Google Colab. These platforms provided us with access to powerful computing resources and collaborative environments where we could experiment with different models. We learned how to prepare datasets, implement model training, monitor performance, and evaluate results. By practicing on Kaggle and Colab notebooks, we also understood how to manage experiments, optimize hyperparameters, and make our models more efficient.

Overall, the internship helped us transform our theoretical knowledge into practical expertise. These skills will not only support our future academic projects but also give us an advantage in professional environments where Python programming and machine learning are essential.

3.2 Knowledge Gained in Terms of Theoretical Understanding

During our practical attachment at INSA, we were able to expand our theoretical knowledge in several important areas of computer science and data analysis. The projects we worked on – image forgery detection, opinion mining, and key point detection – required us to connect what we learned at university with new concepts from research papers and real-world applications.

1. Understanding of Deep Learning Models

One of the biggest areas of growth for us was learning about deep learning models, especially convolutional neural networks (CNNs). Through the image forgery detection project, we studied how CNNs can capture small details in images, such as noise patterns or inconsistencies, that help detect whether an image is real or fake. We also explored how different models are structured, how they pass information through multiple layers, and how this architecture affects their performance.

2. Natural Language Processing (NLP)

From the opinion mining project, we learned a lot about natural language processing (NLP). Before, we mostly thought of text as plain words, but through this project we realized how much effort is required to make text understandable for machines. We studied how tokenization, stemming, and stop-word removal help prepare text, and how these steps are critical for building accurate models. Working on Amharic text gave us new theoretical knowledge about the challenges of processing local languages. Since tools for Amharic are not as developed as for English, we had to read research papers and understand methods researchers used to handle this issue. This taught us how theory often must adapt to context, especially when dealing with low-resource languages.

3. Datasets and Data Preparation

We also improved our understanding of datasets and their importance. We learned that a model's accuracy is not only determined by its design, but also by the quality and balance of the data it is trained on. Cleaning datasets, removing duplicates, and organizing data were tasks we did repeatedly, and this gave us a deeper appreciation of the role data plays in any machine learning project.

4. Evaluation Metrics

Another theoretical area we strengthened was evaluation metrics. By reading research papers, we learned how accuracy alone is not always enough to judge a model. Metrics such as precision, recall, F1-score, and AUC gave us a better understanding of how to measure performance in a meaningful way. These concepts helped us see why some models were better suited for certain tasks and how to fairly compare different approaches.

5. Machine Learning Principles

Overall, all three projects improved our theoretical knowledge of machine learning. We learned how a project goes from research (understanding the theory) to data preparation, to model selection, training, and finally evaluation. This end-to-end process gave us a clear picture of how machine learning works in practice.

6. Research-Oriented Thinking

Finally, reading and analyzing research papers taught us to think more like researchers. We practiced evaluating which models work best, which datasets are more reliable, and what gaps still exist in the field. This helped us bridge the gap between classroom theory and real-world applications in cybersecurity, AI, and data analysis.

3.3 Interpersonal Communication Skills Gained

One of the most valuable aspects of our three-month placement at INSA was the opportunity to **develop strong interpersonal and communication skills**. While the technical projects sharpened our abilities in AI and machine learning, it was the human interactions — with experts, mentors, and fellow students — that truly shaped our professional growth.

At the start, the challenge of communicating with **senior professionals in the field** felt intimidating. We were regularly expected to present research findings, progress reports, and demo results to experts such as **Mr. Tewodros Legesse**, a senior data analytics researcher, and **Dr. Seble Hailu**, an experienced specialist in machine learning classification. Standing in front of such accomplished individuals naturally brought nerves. There was always the thought: *"What if I cannot explain this well enough? What if I get stuck on a question?"* These situations pushed us out of our comfort zone.

However, with every interaction, our confidence grew. Preparing PowerPoint presentations on complex research papers taught us how to **translate technical ideas into clear, structured explanations.** Answering tough questions during feedback sessions strengthened our ability to **think critically under pressure** and defend our choices with evidence. Over time, the nervousness transformed into composure. We began to see communication with experts not as a test, but as a chance to **learn, exchange ideas, and showcase our progress.**

At the same time, the program required us to **socialize and collaborate with students from different universities across Ethiopia.** This experience was equally enriching, as each student brought unique strengths, backgrounds, and approaches to problem-solving. Initially, differences in perspectives and working styles made collaboration challenging. Some students preferred fast decision-making, while others took time to analyze details. Learning how to **adapt communication styles** — whether through simplifying explanations, listening more actively, or mediating disagreements — became a daily exercise in teamwork.

Beyond the work, informal social interactions also mattered. Sharing lunch breaks, debating ideas after group meetings, or simply exchanging stories about our universities helped build bonds that made collaboration smoother and more enjoyable. These moments showed us that effective communication goes beyond formal presentations; it is also about **building trust, empathy, and camaraderie.**

Through these combined experiences, we developed communication skills in three important dimensions:

1. **Professional confidence** – learning to interact with senior experts, overcome nerves, and present ourselves with clarity and credibility.
2. **Team collaboration** – working with students from different universities, balancing diverse perspectives, and achieving collective goals.
3. **Adaptability** – shifting between formal communication with mentors and informal, social communication with peers, while maintaining professionalism in both.

By the end of the placement, the same situations that once created anxiety — whether presenting to a panel of experts or working with new colleagues — had become opportunities to demonstrate growth. The **confidence, patience, and adaptability** gained through these interactions will remain just as important in our future careers as the technical skills we developed.

3.4 Teamwork Skills

During our internship, teamwork was one of the main areas where we improved a lot. Working together taught us how to cooperate, solve problems, and support each other.

One big teamwork experience was when we debugged code as a group. Whenever we faced hard errors or unexpected results, we worked step by step to find the problem. Each of us shared different ideas, and by thinking together we usually solved the bugs faster than working alone. This showed us how teamwork makes problem-solving easier and less stressful.

We also shared ideas and helped each other whenever someone got stuck. For us, teamwork was not only about dividing tasks but also about supporting one another. We gave our own coding ideas and also listened to others. This created an open space where we could learn from each other and use our different strengths.

From these experiences, we learned that good teamwork brings better results and also helps us grow personally. It taught us patience, adaptability, and respect for each other's opinions. We believe these teamwork skills will help us in future school projects and in our professional work.

3.5 Leadership Skills

Working as a group of three gave us many chances to practice and improve our leadership skills. We learned how to share tasks fairly by dividing the work into smaller parts and making sure each person took responsibility for specific modules.

When we faced problems, we practiced problem-solving leadership by discussing possible solutions together. Often, one member would guide the testing process while the others supported, which taught us how to lead and follow depending on the situation.

We also practiced decision-making as a team. From selecting tools and datasets to choosing reporting styles, we made collective choices that required clear communication and mutual respect.

Another important skill was collaboration and motivation. Each member took turns encouraging the group, keeping up the spirit, and reminding everyone about deadlines. This helped us finish our tasks on time despite challenges.

Overall, the group work strengthened qualities like initiative, responsibility, teamwork, and communication — skills that are essential for leadership in any future career.

3.6 Understanding of Work Ethics

Another key outcome of our three-month engagement at INSA was the opportunity to gain a deeper understanding of **work ethics and professional responsibility**. Unlike classroom settings, working in a national research institution required us to adapt to standards and practices that reflect the seriousness of the agency's mission.

1. Punctuality and Time Management

One of the first lessons we experienced was the importance of **punctuality**. At INSA, meetings, presentations, and training sessions began on time, and being late could disrupt not only our progress but also the work of others. Learning to plan ahead, arrive early, and manage our schedules effectively was a key part of developing a strong work ethic. We saw firsthand how respecting time is not just a personal responsibility but also a sign of respect for colleagues and supervisors.

2. Discipline and Accountability

Working within a structured division also showed us the importance of **discipline in completing tasks**. Every stage of our projects — from studying research papers to debugging code — required consistency and commitment. We were held accountable for our work during weekly reviews, which taught us that good work ethics mean taking responsibility for both successes and mistakes. This accountability helped us become more careful, organized, and methodical in the way we approached tasks.

3. Confidentiality and Integrity

Because INSA operates in areas related to **national security and data protection**, we came to understand the ethical responsibility of **confidentiality.** Handling sensitive datasets or discussing research findings required us to be discreet and professional, ensuring that information was not misused or shared inappropriately. This reinforced the value of **integrity** in research and professional work — knowing that trust can only be built when one consistently acts responsibly.

4. Respect and Professional Conduct

Our placement also emphasized the importance of **respect in the workplace**. We had to interact with senior researchers, supervisors, and fellow students with professionalism, regardless of differences in experience or background. This meant active listening during feedback sessions, addressing mentors formally, and giving equal value to contributions from peers. We learned that respectful communication builds a collaborative environment where ideas can be shared freely and productively.

5. Teamwork and Shared Responsibility

Work ethics also revealed themselves in the context of **team collaboration**. We realized that completing large, complex projects was impossible without shared responsibility. When one member delayed or neglected their role, the entire group was affected. This made us more conscious of contributing fairly, supporting peers when needed, and recognizing that strong work ethics are not just individual but collective.

6. Professional Growth Mindset

Finally, we learned that work ethics extend beyond rules — they are also about cultivating the right **attitude toward learning and improvement.** When facing challenges such as debugging errors or presenting research to experts, we were encouraged to persist, seek guidance, and improve with each attempt. This reinforced the idea that professionalism is not about being perfect, but about demonstrating **commitment, honesty, and continuous growth.**

Summary of Lessons that we Learned

From punctuality and accountability to confidentiality and teamwork, our placement helped us build a clearer understanding of the ethical values that sustain professional environments. We realized that in an institution like INSA — where the work directly affects government institutions, critical infrastructure, and citizens — **work ethics are not optional, but essential.**

These lessons will remain central to our future careers, reminding us that technical skills must always go hand in hand with **discipline, responsibility, integrity, and respect.**

3.7 Entrepreneurship Skills

During our internship, we not only learned technical and ethical lessons but also developed an entrepreneurial way of thinking.

We got inspiration to see how AI can solve real problems. By working on data analysis projects, we saw how artificial intelligence can be used in areas like security, healthcare, and automation and other areas. This made us think about how we could use the same ideas in our own future projects or even in possible startups.

We also learned about innovation through data analysis. We realized that new ideas often come from looking at data in different ways, finding hidden patterns, and building solutions that others may not see. This encouraged us to think more creatively and to see problems as chances to create useful solutions.

Through these experiences, we developed an entrepreneurial mindset that will help us in the future—whether we start our own projects or bring new and creative ideas to the organizations we will work with.

Chapter Four: Conclusion and Recommendations

4.1 Conclusion

Our practical attachment at INSA was a truly valuable experience that gave us the chance to connect what we learned in theory with real-world practice. By working on the Image Forgery Detection and Opinion Mining projects, we not only developed stronger technical skills in machine learning, deep learning, and NLP, but also gained confidence in handling datasets, evaluation metrics, and model selection.

Beyond the technical growth, this journey improved our teamwork, leadership, and problem-solving skills. We learned how to face challenges with resilience, share responsibilities effectively, and support each other to meet deadlines. This attachment became a stepping stone that prepared us both academically and professionally for future roles in AI and cybersecurity.

4.2 Recommendations for INSA

Structured Intern Training – Starting with a short orientation about tools, workflows, and expectations would help interns adjust faster and work more efficiently.

Better Computing Resources – Since projects like forgery detection require heavy computation, providing access to high-performance machines or cloud resources would make the work smoother.

Clear Documentation – Having updated project guidelines and manuals would reduce confusion and make it easier for new interns to continue or expand on existing projects.

Cross-Department Exposure – Allowing interns to observe and learn from other areas such as digital forensics, threat analysis, or policymaking would give them a broader view of INSA's work.

Ongoing Mentorship – Regular feedback and check-ins from mentors would guide interns, help them improve continuously, and make the learning process more meaningful.

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